CYSEC

HACKNYX

SHARING SESSION

DOCKER 101 : THE SHIPPING CONTAINER OF APPLICATIONS





Ever heard of Docker but never know how it works?
We'll dive in how does a Docker works and what is even a "Docker" in the first place.

And why is it so widely used by developers and CTF Challenge Creators in hosting a challenges. At the end we will also have a practical session in hosting a website with docker to better understand how does Docker works in our operating system.

WHOAMI?

GNU nano 8.4 introduction.json *

```
{
  "name": "Hariz a.k.a H4rZ",
  "role": "Cybersecurity Student & HackNyx Trainer",
  "study":{
    "institution": "German-Malaysian Institute",
    "program": "Diploma in Cyber Security",
    "interests": [
      "Ethical Hacking",
      "Capture The Flag (CTF)",
      "Red Teaming",
      "Web Exploitation"
    ]
  },
  "roles_in_community": [
    "Technical Unit @  GMI Cyber Security Club",
    "HackNyx CTF Trainer",
    "Cerberix Chairman & Trainer"
  ],
  "socials": {
    "LinkedIn": "Abdul Hariz Mohd Sarnudin",
}
```

WHAT IS DOCKER ?

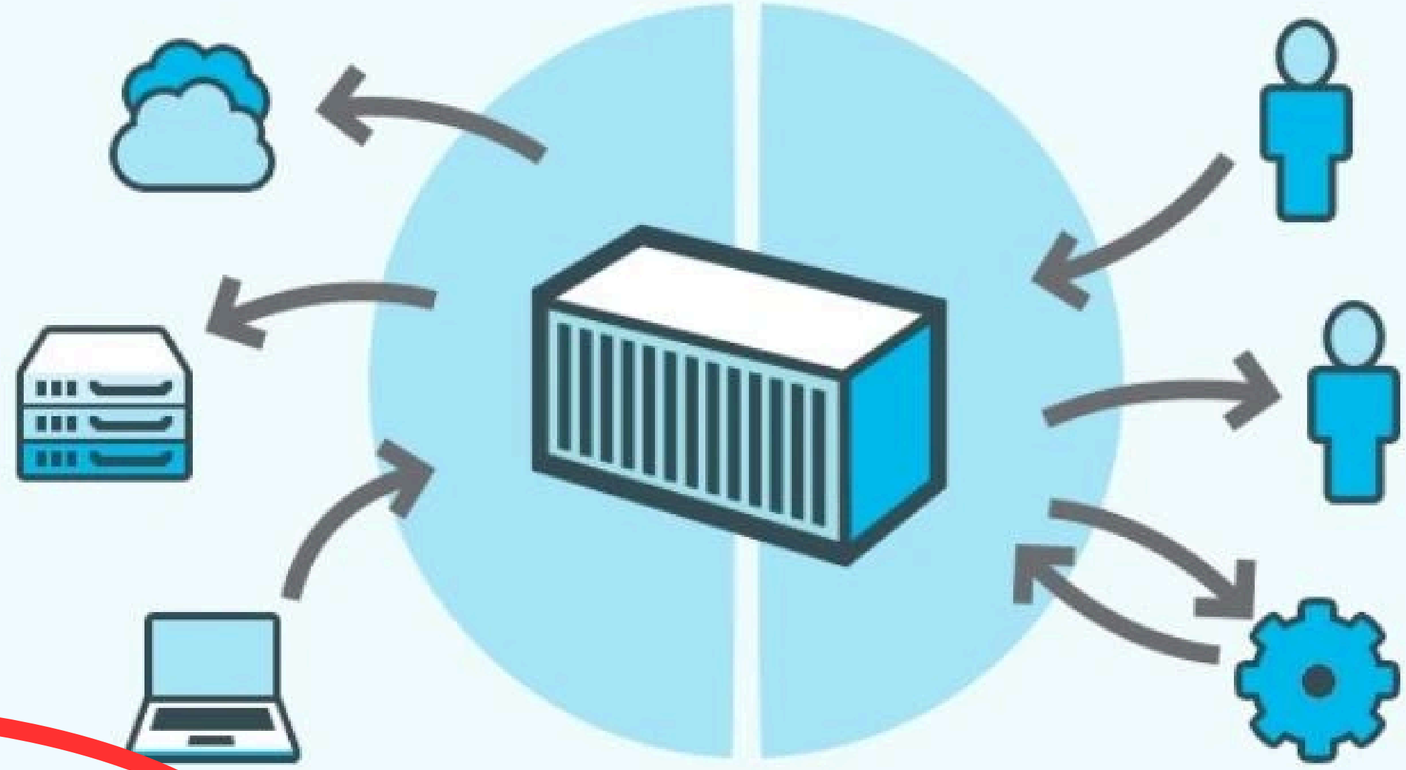


QUICK ANALOGY



What Is Docker?

An open platform for distributed applications



Docker Engine

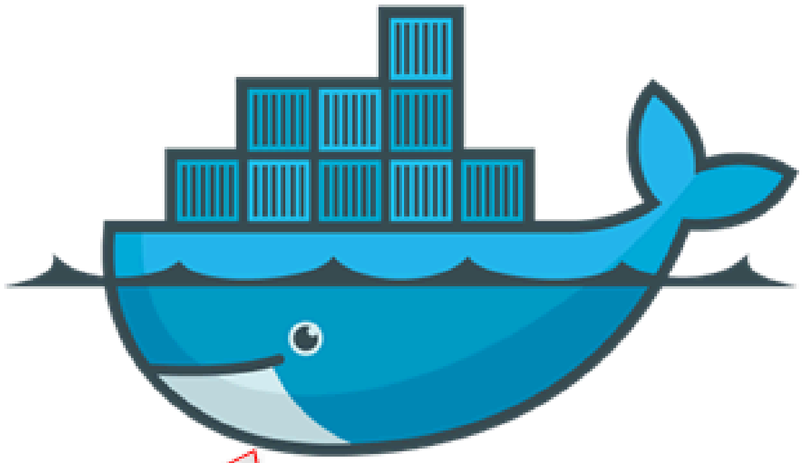
A portable, lightweight application runtime and packaging tool.

Docker Hub

A cloud service for sharing applications and automating workflows.

What is Docker?

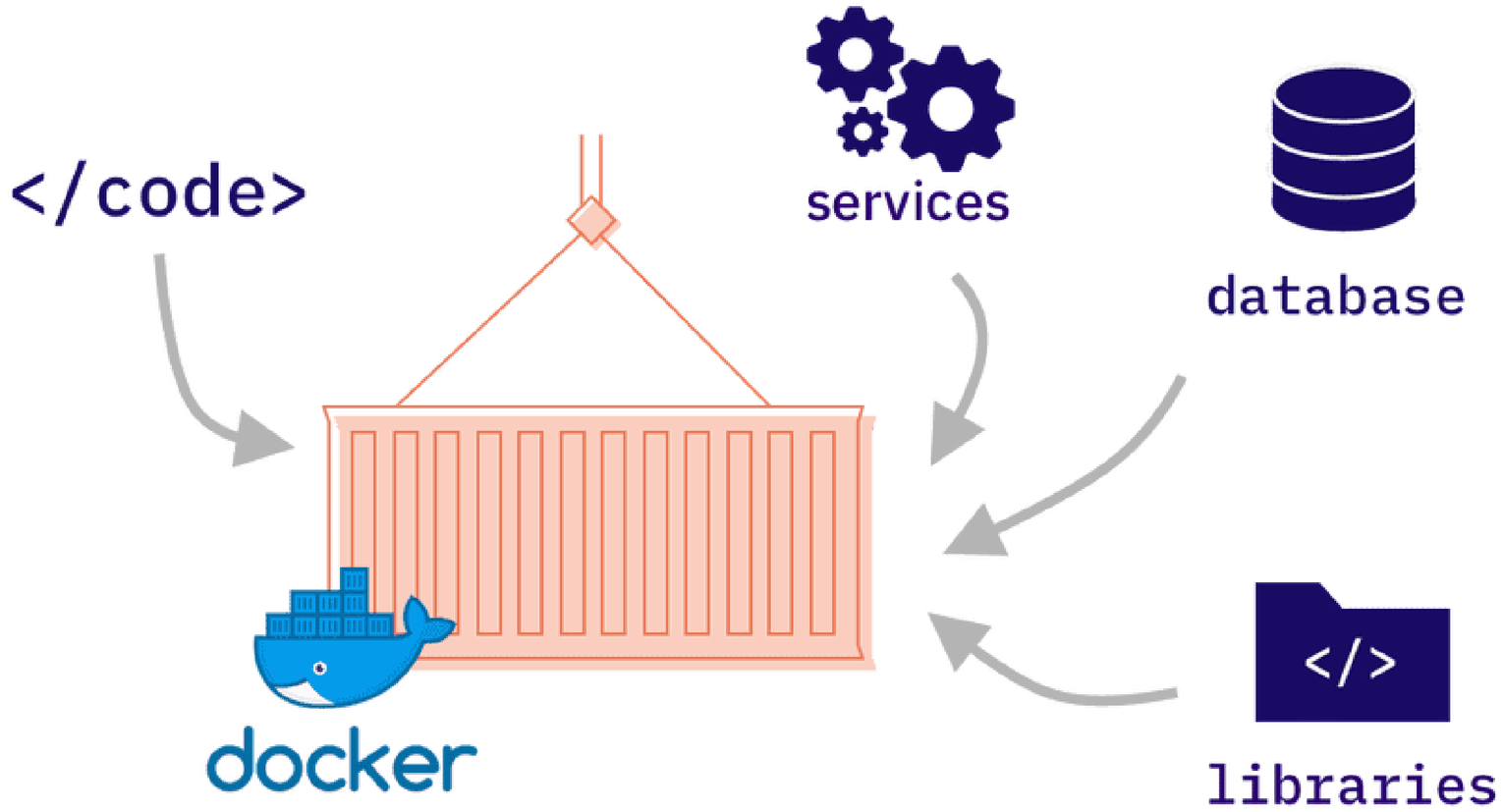
Linux OS isolation tools made easy



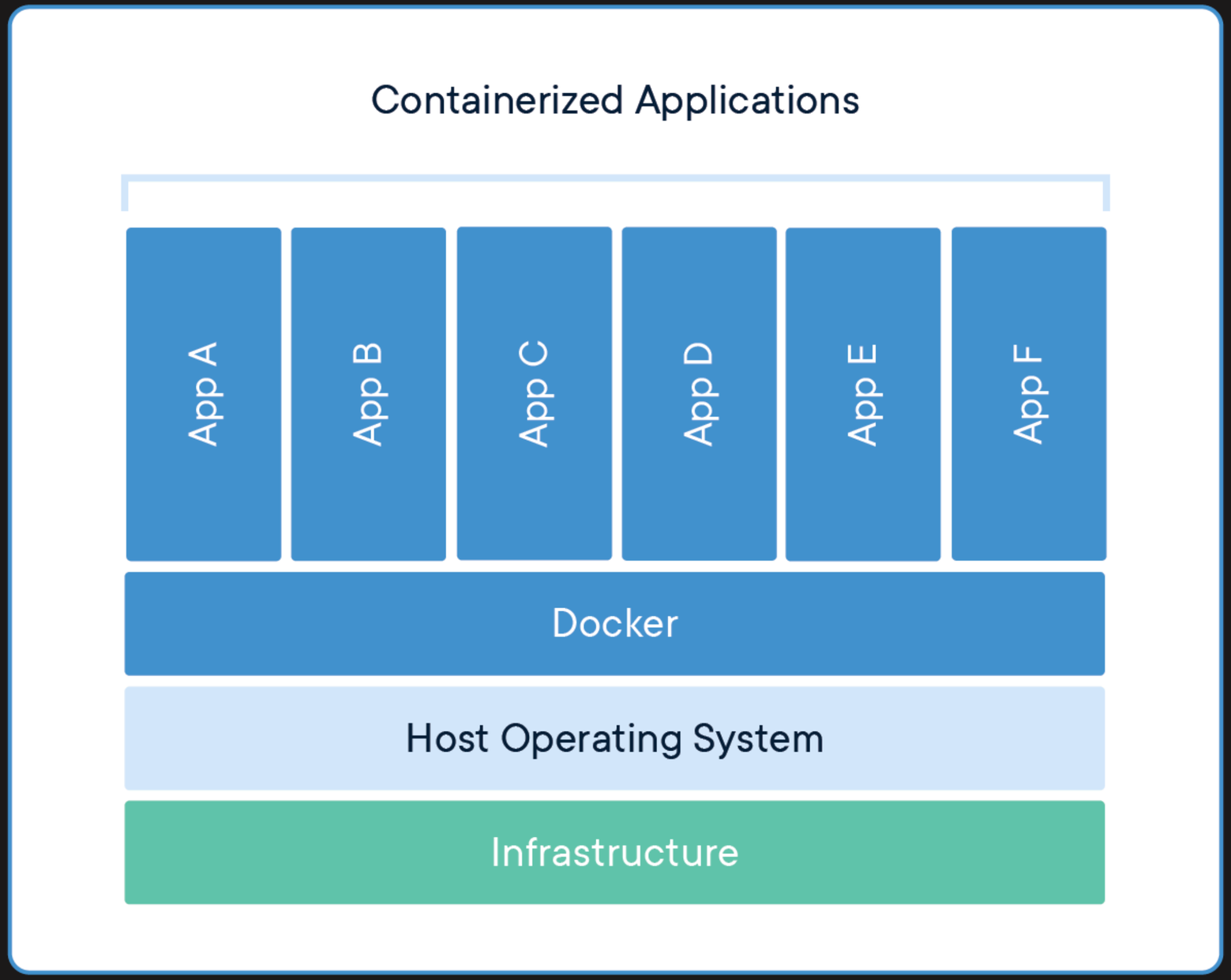
A Docker Container looks like a virtual machine

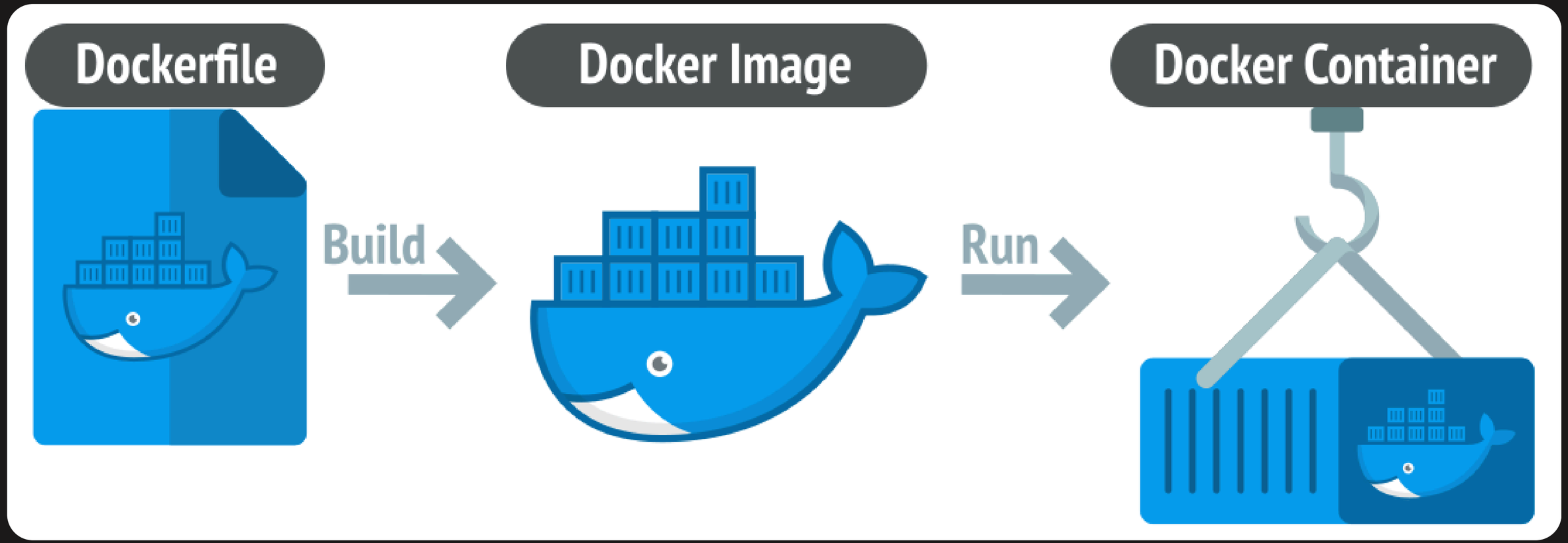
Provide additional 'goodies' for app development

HOW DOES IT WORKS ?

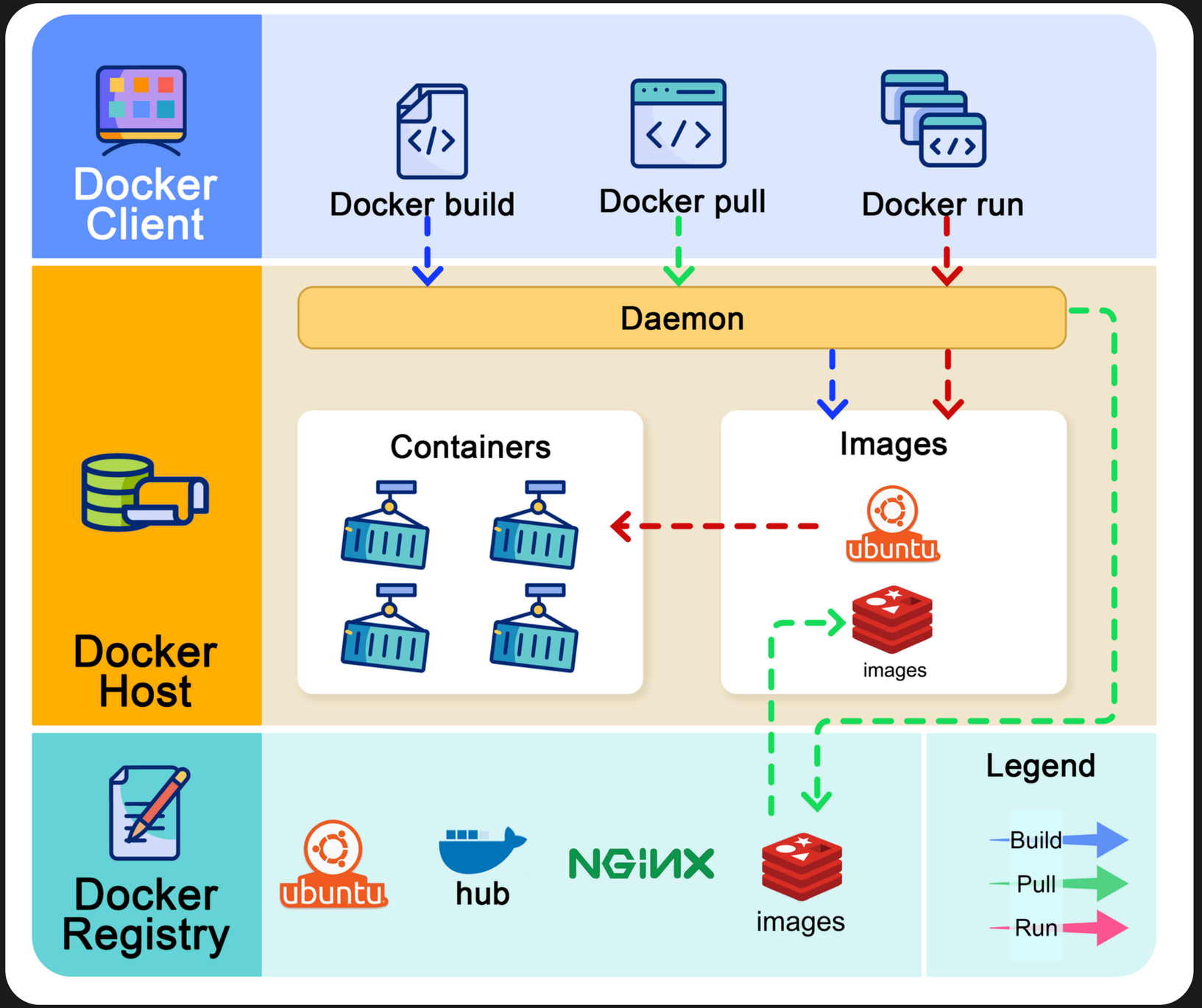


WHAT IS CONTAINER ?




Dockerfile
Docker Image
Docker Container
Build
Run

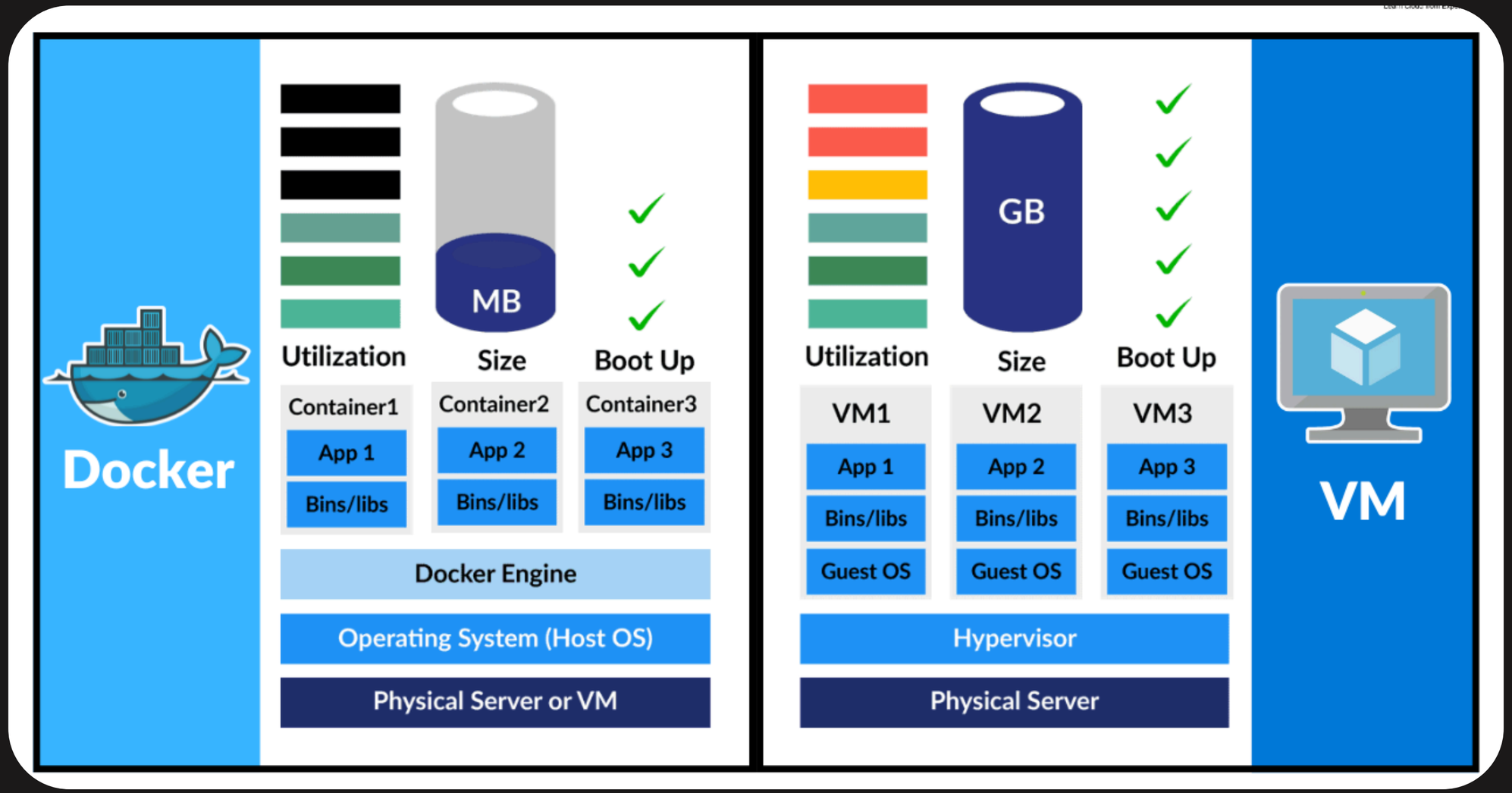
MORE IN DEPTH..



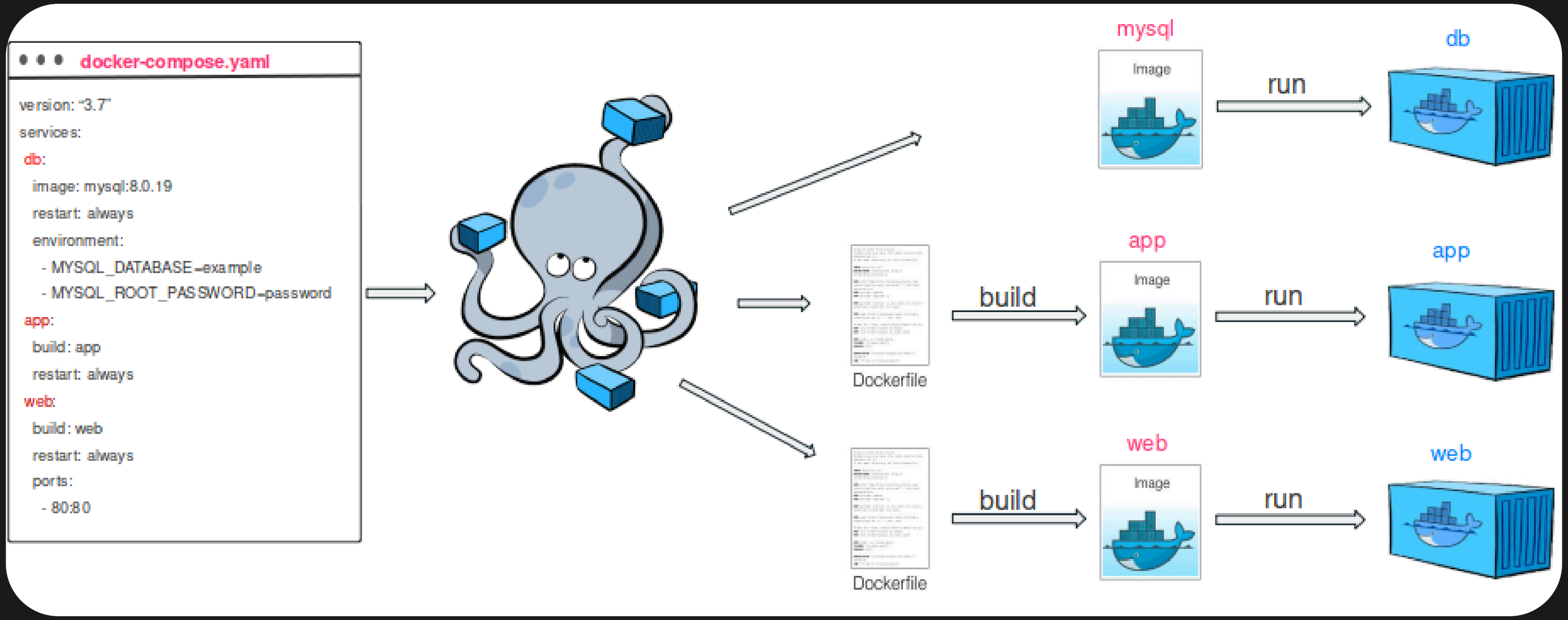
IS IT THE SAME AS VIRTUAL MACHINE ??



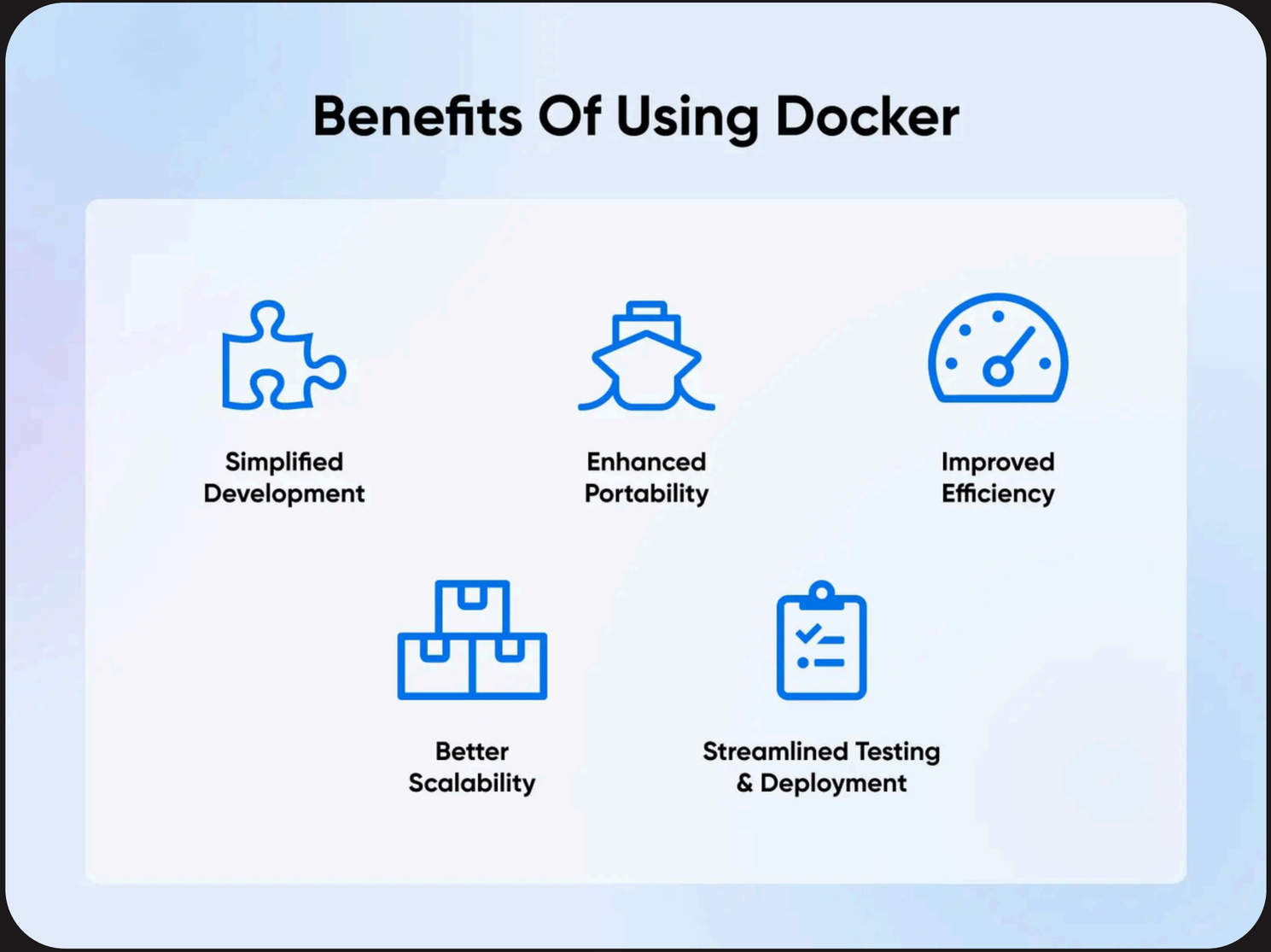
IN A WAY, KIND OF SAME? BUT LIGHTER



DOCKER'S RECIPE BOOK = DOCKER-COMPOSE



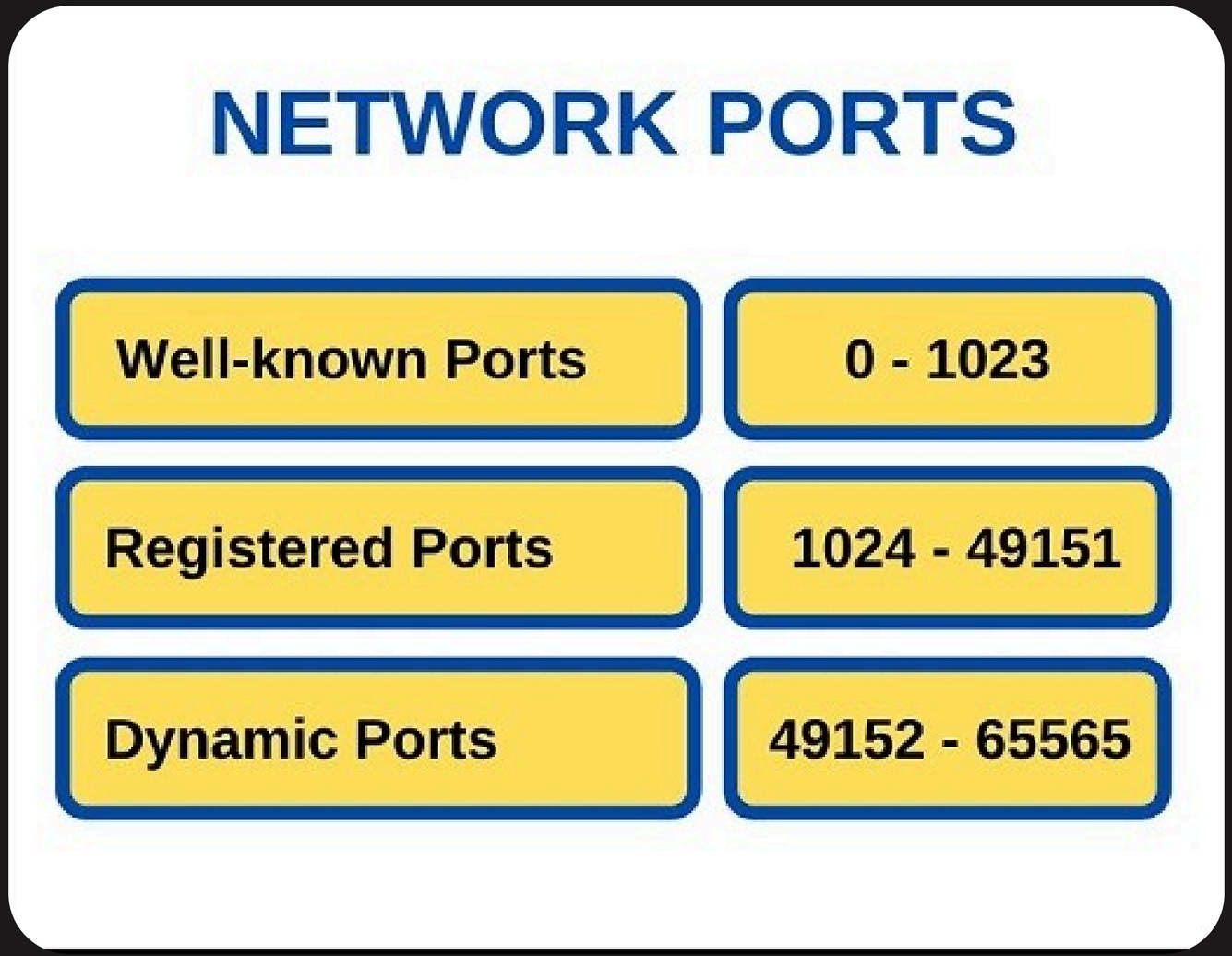
WHAT ARE THE ADVANTAGES ? STOP THE YAPP



HOW TO DEPLOY A WEBTSITE USING DOCKER IN SECONDS.. NO CAP

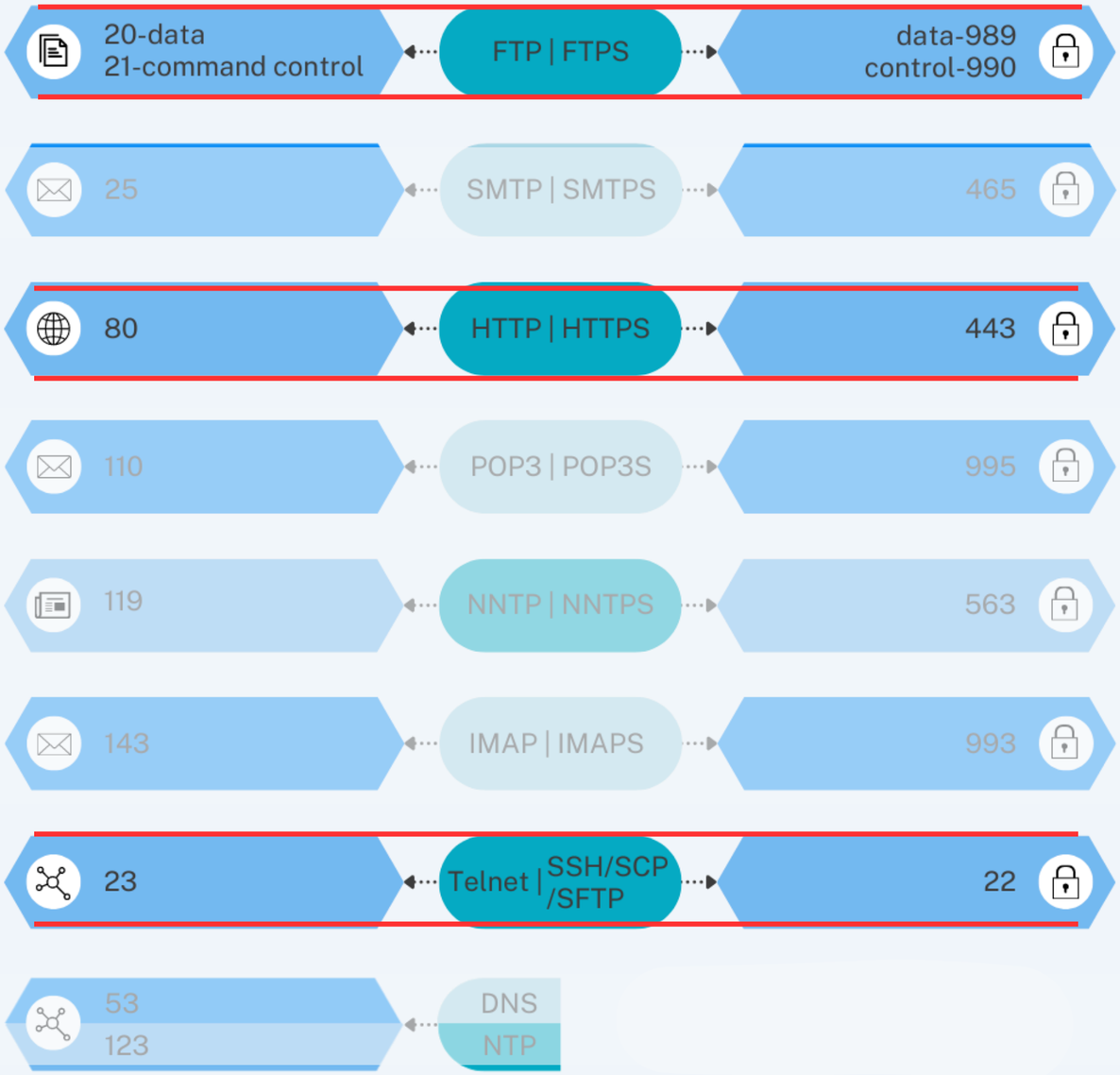
https://youtu.be/zpztxfmcfmg?si=rb4jpvbwj52krnmd&t=296

PORTS & SERVICES



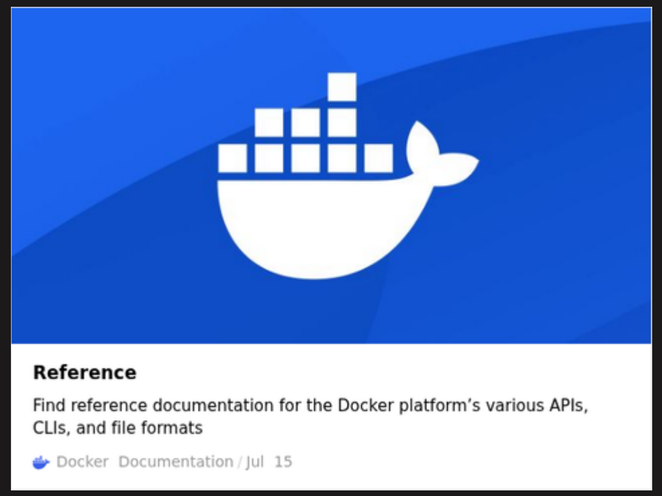
Well-Known Ports: Unencrypted vs Encrypted

Must-know commonly used ports to memorize



EXTRA FOR THOSE WHO WANT TO SELF-LEARN

ALL-LINUX COMMANDS



Reference

Find reference documentation for the Docker platform's various APIs, CLIs, and file formats

Docker Documentation / Jul 15

WINDOWS COMMANDS

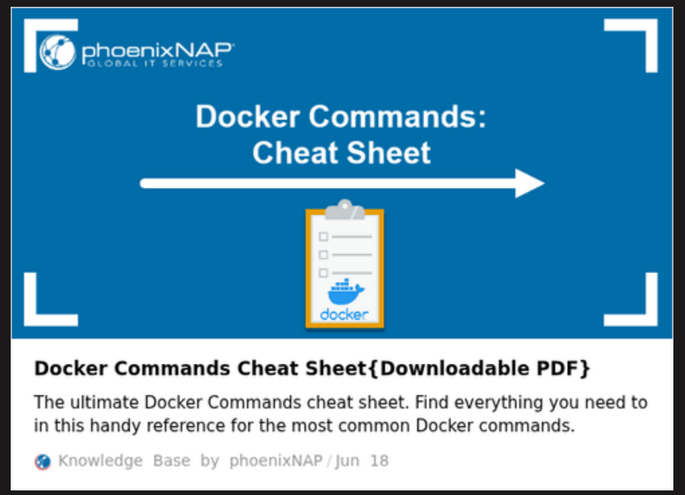


Docker Desktop CLI

How to use the Docker Desktop CLI

Docker Documentation / Aug 19

EXTERNAL SOURCE CHEAT SHEET



Docker Commands Cheat Sheet{Downloadable PDF}

The ultimate Docker Commands cheat sheet. Find everything you need to in this handy reference for the most common Docker commands.

Knowledge Base by phoenixNAP / Jun 18

PRACTICAL

(REFER GIVEN TASKS)



THANK YOU

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